

A Prime Cyclic Cayley Graph: A Disproof of Conjecture 00000003964

GitHub: gaochengzhecpu

Abstract

The source quantifies over every finite simple group, without excluding cyclic groups of prime order. The Cayley graph of $\mathbb{Z}/1021\mathbb{Z}$ with steps ± 1 has diameter at least 510, whereas the proposed upper bound is less than 300. The counterexample works with either natural or binary logarithms. Its simplicity, generation, actual graph-distance lower bound and logarithmic comparisons are verified in Lean.

We use the standard definition of a simple group: a nontrivial group with no proper non-trivial normal subgroup. In particular, a cyclic group of prime order is simple. Both languages of the source say “every finite simple group”; neither imposes a nonabelian hypothesis. For an additive group generated by S , we use the undirected Cayley graph with edges between distinct u, v when $u - v \in S$ or $v - u \in S$. Thus its graph distance is the word metric allowing generators and their inverses.

Theorem 1. *Let $G = \mathbb{Z}/1021\mathbb{Z}$ and $S = \{1\}$. Then G is a finite simple group, S generates G , and its Cayley graph Γ satisfies*

$$\begin{aligned}\text{diam } \Gamma &\geq 510 > 3(\ln |G|)^2, \\ \text{diam } \Gamma &> 3(\log_2 |G|)^2.\end{aligned}$$

Proof. The primes below 32 are

$$2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31.$$

None divides 1021. Since $\sqrt{1021} < 32$, the number 1021 is prime. Hence G has prime order and is simple. Every residue is a nonnegative integer multiple of 1, so S generates it. Its undirected Cayley graph is the cycle on residues $0, 1, \dots, 1020$.

For the representative $a \in \{0, \dots, 1020\}$ of a vertex, set

$$f(a) = \min\{a, 1021 - a\}.$$

Along each edge the potential changes by at most one. Indeed consecutive representatives change each of the two arguments of the minimum by at most one; the wraparound edge joins 1020 and 0, whose potentials are 1 and 0. For a walk of length ℓ from u to v , induction gives $f(v) \leq f(u) + \ell$. Since $f(0) = 0$ and $f(510) = 510$, every walk from 0 to 510 has length at least 510. The graph is connected, so its actual shortest-path distance, and hence its diameter, is at least 510.

Finally, $1021 < 1024 = 2^{10}$ and $0 < \ln 2 < 1$ give

$$0 < \ln 1021 < 10 \ln 2 < 10, \quad 0 < \log_2 1021 < 10.$$

Both proposed upper bounds are therefore strictly less than $300 < 510$. □

Formal verification and scope

The Lean group is the actual type `ZMod 1021`. A kernel-checked primality proof supplies the library's `IsSimpleAddGroup` instance. The theorem `generator_generates` establishes that the actual additive-subgroup closure of $\{1\}$ is the whole group.

The graph is Mathlib's `cycleGraph 1021`, definitionally equal to `circulantGraph ({1} : Set G)`. The library graph symmetrizes the generating set. Connectivity is checked. The potential inequality is proved for every actual edge, extended to every actual walk by induction, and applied to a shortest walk supplied by the connected-graph API. Thus `distance_lower` concerns Mathlib's actual graph distance, not a separately assigned numerical value.

The definition `diameter` is the maximum of these actual distances over all vertex pairs. Lean proves that each distance is at most this maximum, then obtains `diameter_lower`. The theorem `proposed_bound_lt_diameter` checks the strict comparison with the natural-logarithm expression. The additional theorem `binary_bound_lt_diameter` checks the quotient $\ln |G| / \ln 2$ as well. Finally, `conjecture_false` negates the universal diameter upper bound for finite simple groups, written in additive notation, with a generating-set hypothesis and connectivity.

This refutes the source as stated. It does not address a different conjecture restricted to nonabelian finite simple groups. No assertion about the source's additional $\text{PSL}(2, p)$ extremal-family clause is needed, because its universal upper bound already fails.

Source and reproduction

The exact bilingual source is included byte-for-byte as `SOURCE.md`; its repository path is `conjectures/00000003964.md`. The project pins Lean 4.19.0 and Mathlib commit

```
c44e0c8ee63ca166450922a373c7409c5d26b00b.
```

From `lean/`, run `lake build`, then

```
lake env lean -DwarningAsError=true Main.lean.
```

The included public-Git manifest locks all dependency revisions. The `verification/` directory records actual fresh-build logs, theorem-axiom reports, artifact hashes, review and final PDF inspection. No supplementary numerical computation is required.